

Application of a novel optimal control algorithm to a benchmark fed-batch fermentation process

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A novel algorithm for solving nonlinear discrete time optimal control problems with model-reality differences is presented. The technique uses Dynamic Integrated System Optimisation and Parameter Estimation (DISOPE) which has been designed to achieve the correct optimal solution in spite of deficiencies in the mathematical model employed in the optimisation procedure. A method based on Broyden's ideas is used for approximating some derivative trajectories required. Ways for handling constraints on both manipulated and state variables are described. Further, a method for coping with batch-to-batch dynamic variations in the process, which are common in practice, is introduced. It is shown that the iterative procedure associated with the algorithm naturally suits applications to batch processes. The algorithm is successfully applied to a benchmark problem consisting of the input profile optimisation of a fed-batch fermentation process.

Keywords: Batch processes; optimal control; non-linear systems; model uncertainties

1. Introduction

An industrial process is considered to be batch if it consists of a sequence of steps or phases that must be carried out in a defined order (Rijnsdorp, 1991). The culmination of this succession of steps creates a finite amount of finished product. The sequence needs to be repeated to produce additional amounts of product. Batch processes present interesting control problems due to their inherent dynamic nature.

Batch processes have optimum conditions that will yield the maximum product with minimum time and cost. The degrees of freedom for the determination of such optimum conditions are often a combination of the initial conditions, the set-point profile used during the transformation phase and the time allowed for this phase. Procedures for determining acceptable set-point profiles include trial and error, previous experience, and the use of dynamic optimisation.

Optimal set-point profiles obtained by the use of optimal control theory based on a dynamic model of the batch process will only be optimal for the specific model and parameter values used in the optimisation. This mismatch between the model and the actual plant

behaviour may result in sub-optimal performance when the model-based optimal profile is applied to the real process.

The ISOPE algorithm, which is an acronym for Integrated System Optimisation and Parameter Estimation, was originally developed by Roberts (1979) for on-line steady-state optimisation of industrial processes. The algorithm is implemented through adjustment of regulatory controller set-points where these set-points are calculated by using a steady-state model of the process. An important property of the ISOPE algorithm is that the iterations converge to the real optimum in spite of model-reality differences between the real process and the model employed. In further research (Roberts, 1992, 1993), the ISOPE principle was extended to develop an iterative technique for solving continuous time dynamic optimal control problems which takes into account that the model used for the computations is different from reality. This gave rise to the continuous time DISOPE algorithm (Dynamic Integrated System Optimisation and Parameter Estimation).

In industrial practice, many control schemes are implemented digitally where the control input is switched between different values at discrete time steps. The control signal is normally held constant between samples by a zero-order hold. Such controls are usually designed using a discretised version of the continuous plant. There are also processes which are discrete in nature and can only be controlled by using discrete time controllers. For these types of problems a discrete formulation of DISOPE has been developed, analysed and implemented (Becerra and Roberts, 1996a; Becerra, 1994).

In this work the principles behind the discrete version of DISOPE are described and the algorithm is shown to be particularly applicable to batch processes because of their inherent repetitive nature. By matching the iterations of the DISOPE algorithm appropriately to the batch cycles it is shown that this is a natural application for DISOPE in which its inherent advantage for coping with model-reality differences can be utilised profitably.

In this paper, the work presented by Becerra and Roberts (1995, 1996b) is extended to cope with batch-to-batch random variations in the initial conditions of the process and with batch-to-batch variations in the dynamic behaviour of the process. A method based on Broyden's ideas (Broyden, 1965), is used to compute some first derivative trajectories required by DISOPE. The technique is applied to a benchmark problem

consisting of the input profile optimisation of a fed-batch fermentation process. Simulation results are presented showing improvements compared with some previously known nominal feed profiles.

This paper is organised as follows: An introduction to the DISOPE technique, its formulation and its application to batch process optimisation, are presented in Section 2.1. A description of the method for approximating some Jacobian trajectories which are required is given in Section 2.2. The methods used for handling constraints and batch-to-batch dynamic variations are presented in Sections 2.3 and 2.4, respectively. Comparisons with other methods for batch process optimisation are given in Section 2.5. In Section 3.1 the fed-batch fermentation process under study is briefly described. The identification method used for finding an approximate linear state space model of the process is described in Section 3.2.1. The optimal control design parameters and results are presented and analysed in Section 3.2.2. Critical comments and further research topics are given in Section 4. Finally, concluding remarks are given in Section 5.

2. The DISOPE algorithm

2.1 Formulation

Consider the following real optimal control problem (ROP), which is formulated in discrete time

$$\min_{u(k)} J = \varphi^*(x(N)) + \sum_{k=0}^{N-1} L^*(x(k), u(k), k)$$

$$\begin{aligned} \text{s.t. } x(k+1) &= f^*(x(k), u(k), k); k \in [0, N-1] \\ x(0) &= x_0 \end{aligned} \quad \dots (1)$$

where $u(k) \in \mathbb{R}^m$ and $x(k) \in \mathbb{R}^n$ are the discrete control and state vectors, respectively, $\varphi^*: \mathbb{R}^n \rightarrow \mathbb{R}$ is called the real terminal measure, $L^*: \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R} \rightarrow \mathbb{R}$ is the real performance measure function and $f^*: \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R} \rightarrow \mathbb{R}^n$ represents the real process dynamics. Instead of solving the ROP, the following modified model-based optimal control problem (MMOP) is solved (for full details see Becerra and Roberts, 1996a; Becerra, 1994):

$$\begin{aligned} \min_{u(k)} J &= \varphi(x(N), \gamma_1) - \Gamma_1^T x(N) + \sum_{k=0}^{N-1} [L(x(k), u(k), \gamma_2(k)) \\ &\quad - \lambda(k)^T u(k) - \beta(k)^T x(k) \\ &\quad + \frac{1}{2} r_1 \|u(k) - v(k)\|^2 + \frac{1}{2} r_2 \|x(k) - z(k)\|^2] \\ \text{s.t. } x(k+1) &= f(x(k), u(k), \alpha(k)); \quad k \in [0, N-1] \\ x(0) &= x_0 \end{aligned} \quad \dots (2)$$

where $\varphi: \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}$ is called the model terminal measure, $L: \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R} \rightarrow \mathbb{R}$ is the model performance measure function and $f: \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^r \rightarrow \mathbb{R}^n$ represents the dynamic model. The parameters $\alpha(k) \in \mathbb{R}^r$, $\gamma_2(k) \in \mathbb{R}$ and $\gamma_1 \in \mathbb{R}$ are associated with the functions f , L and φ , respectively; $v(k) \in \mathbb{R}^m$ and $z(k) \in \mathbb{R}^n$ are introduced to separate the control and state variables between the so-called optimisation and

parameter estimation problems (to be described below). The sequence $v(k) k \in [0, N-1]$ represents the input profile applied to the batch process along every batch. The terms proportional to r_1 and r_2 are introduced to augment the performance index to provide convexification. This is done to improve convergence in difficult cases.

The solution of MMOP is achieved under specified parameters γ_1 , $\alpha(k)$ and $\gamma_2(k)$, specified multipliers Γ_1 , $\lambda(k)$ and $\beta(k)$, and specified $v(k)$ and $z(k)$; where:

$$\begin{aligned} f(z(k), v(k), \alpha(k)) &= f^*(z(k), v(k), k) \\ L(z(k), v(k), \gamma_2(k)) &= L^*(z(k), v(k), k) \end{aligned} \quad \dots (3)$$

$$k \in [0, N-1]$$

$$\begin{aligned} \lambda(k) &= \left[\frac{\partial f}{\partial v(k)} - \frac{\partial f^*}{\partial v(k)} \right]^T \hat{p}(k+1) + \left[\frac{\partial L}{\partial v(k)} - \frac{\partial L^*}{\partial v(k)} \right]^T \\ \beta(k) &= \left[\frac{\partial f}{\partial z(k)} - \frac{\partial f^*}{\partial z(k)} \right]^T \hat{p}(k+1) + \left[\frac{\partial L}{\partial z(k)} - \frac{\partial L^*}{\partial z(k)} \right]^T \\ k &\in [0, N-1] \end{aligned} \quad \dots (4)$$

$$\varphi(z(N), \gamma_1) = \varphi^*(z(N)) \quad \dots (5)$$

$$\Gamma_1 = \nabla_z \varphi(z(N), \gamma_1) - \nabla_z \varphi^*(z(N)) \quad \dots (6)$$

$$v(k) = u(k); k \in [0, N-1]$$

$$z(k) = x(k); k \in [0, N] \quad \dots (7)$$

$$\hat{p}(k) = p(k); k \in [0, N]$$

with $\hat{p}(k)$ introduced as a separation variable for the co-state $p(k)$ obtained by solving MMOP. The costate variable $p(k)$ and its separation variable $\hat{p}(k)$ arise from optimisation theory applied to dynamic optimal control problems such as (1). The co-state is the Lagrange multiplier of the state equations, which are treated as equality constraints of the optimisation problem. The need to introduce separation variables for the input profile, state and co-state sequences arises from the need to allow for the interaction between system optimisation and parameter estimation (Becerra and Roberts, 1996a).

Assuming convergence, the discrete time DISOPE algorithm (Becerra, 1994) achieves the necessary optimality conditions of the ROP via repeated solutions of the MMOP. It has been designed so that it handles the differences between the model used for the optimisation calculations and the real process, in such a way that the optimality conditions of ROP are achieved in a sequential way in spite of the deficiencies in the model. The idea of applying DISOPE to the dynamic optimisation of batch processes arises naturally from knowledge on the structure of the algorithm and the basic principles on which it was originally developed. It is possible to integrate the iterations of DISOPE towards the dynamic optimum with the batchwise operation of the process. This is achieved by introducing a modification on the basic algorithm such that the control profile obtained

from model-based optimisation at each iteration is applied to the real process at every transformation phase. Then, assuming that the transformation time is fixed, the discrete-time DISOPE algorithm for batch processes is described as follows:

Algorithm 1: Discrete-time DISOPE for batch processes

Data: f, L, φ, x_0, N and means for calculating f^*, L^* and φ^* .

Step 0: Compute or choose a nominal solution $v^0(k)$, $k \in [0, N-1]$, and $p^0(k)$, $k \in [0, N]$. Set $i = 0$.

Step 1: During the transformation phase, apply the control profile $v^i(k)$ to the batch plant. Obtain the corresponding state response $z^i(k)$, $k \in [0, N]$ and dynamic derivatives $\partial f^*/\partial z(k)$ and $\partial f^*/\partial v(k)$, $k \in [0, N-1]$.

Step 2: Compute the parameters $\alpha^i(k)$, $\gamma_2^i(k)$, $k \in [0, N-1]$, to satisfy (3) and γ_1^i to satisfy (5). This is called the *parameter estimation step*.

Step 3: Compute the multipliers $\lambda^i(k)$ and $\beta^i(k)$, $k \in [0, N-1]$, from (4) and Γ_1^i from (6).

Step 4: With specified $\alpha^i(k)$, $\gamma_2^i(k)$, $\lambda^i(k)$, $\beta^i(k)$, $k \in [0, N-1]$, γ_1^i and Γ_1^i solve the discrete-time modified model-based optimal control problem MMOP to obtain $u^{i+1}(k)$, $k \in [0, N-1]$, $x^{i+1}(k)$ and $p^{i+1}(k)$, $k \in [0, N]$. This is called the *system optimisation step*.

Step 5: This step tests convergence and updates the estimate for the optimal solution of ROP. In order to provide a mechanism for regulating convergence, a simple relaxation method is employed to satisfy (7). This is

$$\begin{aligned} v^{i+1}(k) &= v^i(k) + k_v(u^{i+1}(k) - v^i(k)) \\ \hat{p}^{i+1}(k) &= \hat{p}^i(k) + k_p(p^{i+1}(k) - \hat{p}^i(k)) \end{aligned} \quad \dots (8)$$

where k_v and k_p are scalar gains (usually $\in (0, 1]$). If $v^{i+1}(k) = v^i(k)$, $k \in [0, N-1]$ within a given tolerance stop, else set $i = i + 1$ and continue from step 1.

It is important to note that Algorithm 1 requires knowledge of the values of the state variables of the process during the transformation phase and, furthermore, the values of the derivative matrices $\partial f^*/\partial z$ and $\partial f^*/\partial v$ are also required during that period. The state variables may probably be measured, noticing that on-line measurements in real time are not required. The discrete-time approach allows, for instance, that samples of substances be collected along the transformation phase and sent for laboratory analysis. A method for obtaining estimates of the first derivatives is given below.

2.2 Estimation of first derivatives

Given the (unknown) real process discrete time mapping:

$$x(k+1) = f^*(x(k), u(k)) \quad \dots (9)$$

Define the augmented vector $X = [x \ u]^T$. As a consequence $f^*(x, u) = f^*(X)$. Given two different trajectories

$x^i(\cdot)$, $x^{i+1}(\cdot)$, and $u^i(\cdot)$, $u^{i+1}(\cdot)$, define the change trajectory as:

$$\delta X^i(\cdot) = \begin{bmatrix} x^{i+1}(\cdot) - x^i(\cdot) \\ u^{i+1}(\cdot) - u^i(\cdot) \end{bmatrix} \quad \dots (10)$$

The state and control trajectories denoted by the super-indexes i and $i+1$ may be interpreted as two successive iterates of the DISOPE algorithm.

Using a technique based in Broyden's formula (Broyden, 1965) for approximating the Jacobian matrix in the Quasi-Newton method for solving nonlinear equations, it is possible to define a recursion on the Jacobian trajectories, given two successive control and state trajectories:

$$D^{i+1}(\cdot) = D^i(\cdot) + \frac{[f^{*i+1}(\cdot) - f^{*i}(\cdot) - D^i(\cdot)\delta X^i(\cdot)]\delta X^i(\cdot)^T}{\|\delta X^i(\cdot)\|^2} \quad \dots (11)$$

where $D(\cdot)$ is the $n \times (n+m)$ Jacobian matrix trajectory:

$$\begin{aligned} D(\cdot) &= \frac{\partial f^*(X(\cdot))}{\partial X(\cdot)} \\ &= \left[\frac{\partial f^*(x(\cdot), u(\cdot))}{\partial x(\cdot)} \quad \frac{\partial f^*(x(\cdot), u(\cdot))}{\partial u(\cdot)} \right] \end{aligned} \quad \dots (12)$$

Note that $x(k+1) = f^*(X(k)) = qx(k)$, where q is the forward shift operator. Thus $f^*(\cdot)$ in Eqn (11) is the state trajectory $x(\cdot)$ shifted by one sample.

Eqn (11) is a recursion on the Jacobian trajectories $D(\cdot)$. It uses information of state and control trajectories along the transformation phase of the batch process. This recursion may be coupled to Algorithm 1 in order to provide estimates of the first derivatives required to compute the multipliers in Step 3. An initial value of the trajectory $D(\cdot)$ is required to start the recursion.

2.3 Handling of random batch to batch variations

A typical characteristic of industrial batch processes is the occurrence of random batch-to-batch variations in the initial conditions of the process. In addition, the dynamic behaviour of the process may change from batch to batch. This may be caused by internal parameter variations and also by stochastic disturbances. A consequence of these phenomena is that for the same input profile, the response of the batch process is different from batch to batch.

In the formulation presented in Section 2.1 it has been assumed that the process dynamics are unknown but fixed. A way of handling the random variations described above is to minimise the expected value of the objective function (EJ). This is done in Algorithm 1 by working with the expected value of the state trajectory $\{z^i(\cdot)\}$. In step 1 of Algorithm 1, the expected value of the response of the process $E\{z^i(\cdot)\}$ for a given input profile $\{v^i(\cdot)\}$ may be approximated by an average

of the process response for a finite number of batches $\{z_1^i(\cdot)\} \dots \{z_{N_b}^i(\cdot)\}$:

$$\bar{z}^i(k) = \frac{1}{N_b} \sum_{j=1}^{N_b} z_j^i(k); \quad k \in [0, N] \quad \dots (13)$$

2.4 Handling of constraints

2.4.1 Control magnitude constraints

Bound constraints on the decision variables $u(k)$:

$$u_l \leq u(k) \leq u_h \quad \dots (14)$$

can easily be accommodated within Algorithm 1 by using a variable transformation technique (Becerra, 1994). In this way, the use of a saturation function both when evaluating the function L^* , and when applying the input profile $\{v(\cdot)\}$ to the process, converts this type of constraint into a nonlinearity of the real problem, while the solution of MMOP in step 5 of Algorithm 1 remains unconstrained.

2.4.2 State constraints

State constraints are handled within DISOPE by using a penalty relaxation approach (Becerra, 1994). Assume that s state dependent inequality constraints are defined:

$$\Psi(x(k)) \geq 0 \quad \dots (15)$$

where $\Psi: \mathbb{R}^n \rightarrow \mathbb{R}^s$ is the state constraint function. By using the penalty relaxation technique, the original state constrained problem is transformed into an unconstrained problem by adding a penalty term to the original performance weighting function L^* :

$$L_\rho^*(x(k), u(k)) = L^*(x(k), u(k)) + \rho \sum_{j=1}^s [P_\varepsilon(\Psi_j(x(k)))]^2 \quad \dots (16)$$

where ρ is a penalty factor and $P_\varepsilon: \mathbb{R} \rightarrow \mathbb{R}$ is a smoothed function given by:

$$P_\varepsilon(w) = \begin{cases} w & w \leq -\varepsilon \\ -(w - \varepsilon)^2 / 4\varepsilon & -\varepsilon < w < \varepsilon \\ 0 & w \geq \varepsilon \end{cases} \quad \dots (17)$$

where w is a given scalar argument and ε is a small scalar value. The penalty term becomes significant in the face of violations of any state constraint along the time horizon, while its value is zero if no constraints are violated.

2.5 Comparison with other methods

The dynamic optimisation of batch processes attempts to find the best input profiles during a batch run. Methods for optimising batch processes can be classified in three categories: one-time optimisation, batch-to-batch optimisation and on-line optimisation. See the paper by Bonvin (1997).

(a) *One-time optimisation*: An optimal control problem is formulated on the basis of a dynamic model of the

process. The solution provides the required input trajectories. These profiles are computed off-line and hence are highly sensitive to modelling errors and disturbances.

(b) *Batch-to-batch optimisation*: As additional information becomes available with the completion of each batch run, this information is used to improve future operations. The calculations required by these methods are normally carried out in the intermediate period between two consecutive batch runs. The algorithm presented in this paper belongs to this category. A related batch-to-batch method was proposed by Zafiriou and Zhu (1989). Their method uses a process model to compute input profiles, costates and first derivatives along the trajectories, and uses the plant itself to generate the required state trajectories. A disadvantage of their method is the use of the process model for estimating the derivatives, which makes the resulting profiles highly sensitive to modelling errors. In the method presented in this paper it is also assumed that the states are measured but an attempt is made to use measured information from the process to obtain estimates of the real dynamic derivatives.

(c) *On-line optimisation*: These methods try to compensate the fact that in the presence of modelling errors and disturbances the input profiles computed off-line become sub-optimal. This is done by repeating on-line model-based optimisation several times during a batch run using real time measurements, so introducing feedback in the calculation of the input profiles. These methods require computer intensive real-time calculations, a valid model of the process and real-time estimates of the states and parameters.

3. Application to a simulated fed-batch fermentation process

3.1 Process description

The simulated fed-batch fermentation process under study produces a secondary metabolite. The micro-organisms in this process need two substrates. Two feeds are used to provide the substrates to the culture. A nonlinear dynamic model of the process has been provided to the authors in scrambled C code by the organisers of a batch process control competition at the Industrial Control Centre, University of Westminster, London, UK. The model source code is available from the authors. The following phenomena are simulated in the model to give it a realistic behaviour:

- The initial state varies randomly within a subspace for each batch.
- The parameters of the model vary within certain bounds from batch to batch.
- Some unmeasured disturbances.

The process has two inputs:

U_1 : Substrate 1 feed rate
 U_2 : Substrate 2 feed rate

There are five measurements:

X : Biomass
 S_1 : Concentration of substrate 1
 S_2 : Concentration of substrate 2

P : Product concentration
 V : Volume

All variables are dimensionless. The inputs of the process are constrained between the levels 0 and 50. In addition the volume of the fermentor is constrained to be lower or equal than 4000.

The nominal input profiles U_1 and U_2 which have been provided as part of this benchmark problem were:

$$U_1(t) = 10 + \frac{25}{1 + e^{5-0.5t}} \quad \dots(18)$$

$$U_2(t) = 3.5 - \frac{3.5}{1 + e^{10-0.15t}}$$

where t is the time variable.

The purpose of the control design is to achieve maximum productivity, which can be quantified by means of the following productivity measure:

$$J_p = \frac{P(T_b) \times V(T_b)}{T_b} \quad \dots(19)$$

where T_b is the duration of a batch. This productivity measure quantifies the amount of final product (Concentration $\times$ Volume) per unit of batch time.

3.2 Problem solution

3.2.1 Initial model identification

A version of Algorithm 1 has been implemented where the model-based problem defined by the functions L , f and φ , was chosen to be Linear-Quadratic. However, the Real Optimal Control Problem in this case study is based upon nonlinear dynamics. In general, the formulation presented in Section 2.1 allows the use of a general discrete time nonlinear model to form the model-based problem. Although fed-batch fermentation processes are highly nonlinear, for this case study a simple empirically calculated linear state space model of the process was used to define the model-based problem. One computational advantage of using a linear model and a quadratic model-based objective is that standard non-iterative Linear-Quadratic methods may be used for the solution of the model-based problem at each iteration. Notice, however, that DISOPE has been designed such that model–reality differences are appropriately handled. The use of a linear dynamic model is a tougher test for the convergence of Algorithm 1, because probably a nonlinear empirical model should approximate more

closely the dynamics of the nonlinear system which plays the role of the real system in this case study.

The structure of the dynamic equation used in MMOP is as follows:

$$x(k+1) = f(x(k), u(k), \alpha(k)) = Ax(k) + Bu(k) + \alpha(k) \quad \dots(20)$$

where A and B are known matrices which represent a linear approximation of the real and unknown dynamic mapping f^* .

Suitable values of A and B can be identified from input/output data using the Least Squares method. The following ARX model structure with first-order matrix polynomials is assumed:

$$A^*(q^{-1})y(k) = B^*(q^{-1})u(k) + \varepsilon(k)$$

$$A^*(q^{-1}) = I + A_1q^{-1}$$

$$B^*(q^{-1}) = B_1q^{-1}$$

The relationships between the state space matrices and the ARX polynomial coefficients are:

$$A = -A_1 \quad \dots(22)$$

$$B = B_1$$

Each input signal used for the identification experiment was chosen as a constant feed profile plus a pseudo random binary sequence ($U_1 = 5.0 + / - 1.0$; $U_2 = 3.0 + / - 0.3$). The random sequences were added with the purpose of enhancing input excitation, and each of them had a probability of switching values of 0.5 at every sampling time. Numerical scaling was applied to the input/output data set obtained. The following scaling factors, which were chosen as the reciprocal of the mean value of each variable, were used:

$$U_1: 0.20067$$

$$U_2: 0.33278$$

$$X: 0.06567$$

$$S_1: 0.44048$$

$$S_2: 5.81487$$

$$P: 0.0126817$$

$$V: 0.000647604$$

Using Matlab's System Identification Toolbox (The Mathworks, 1995), the following state space model matrices were obtained:

$$A = \begin{bmatrix} 9.4703955e-001 & 1.0808167e-003 & -6.3441285e-003 & 5.6855561e-003 & 1.8371262e-002 \\ -4.1075479e-001 & 9.4441240e-001 & 2.3167529e-001 & 3.5875480e-001 & 2.1383912e+000 \\ -2.5992268e+000 & -2.1650136e-001 & 9.9778871e-001 & -4.4781740e-001 & 5.8478423e+000 \\ -3.2407403e-001 & -3.1749288e-002 & -8.1140282e-003 & 8.6966071e-001 & 4.5745610e-001 \\ 4.7046670e-001 & 1.1621591e-002 & 2.8518379e-003 & -4.9560711e-003 & -1.0937075e-001 \end{bmatrix}$$

$$B = \begin{bmatrix} 3.4805828e-003 & -1.0682101e-002 \\ 4.2571328e-002 & 2.0812828e-001 \\ -7.7516745e-002 & 5.0089347e-001 \\ 6.4665320e-002 & -1.2225967e-002 \\ 8.1323742e-004 & -5.2762750e-003 \end{bmatrix}$$

3.2.2 Optimal control design and testing

Define the state and control vectors as follows:

$$x = [X \ S_1 \ S_2 \ P \ V]^T; \quad u = [U_1 \ U_2] \quad \dots (23)$$

The performance index to be minimised reflects the desire to maximise the amount of product for a given batch time, and is given by:

$$J^* = -x_4(N)x_5(N) \quad \dots (24)$$

where $N = 119$ corresponds with the transformation time of 120 units, given a sampling interval of one time unit, taking 0 as the initial time index of every batch or iteration. The MMOP was based on a linear dynamic model:

MMOP

$$\min_{u(k)} J_m = -\Gamma_1^T x(N) + \sum_{k=0}^{N-1} \frac{1}{2} r_1 \|u(k) - v(k)\|^2 - \lambda(k)^T u(k) - \beta(k)^T x(k) \quad \dots (25)$$

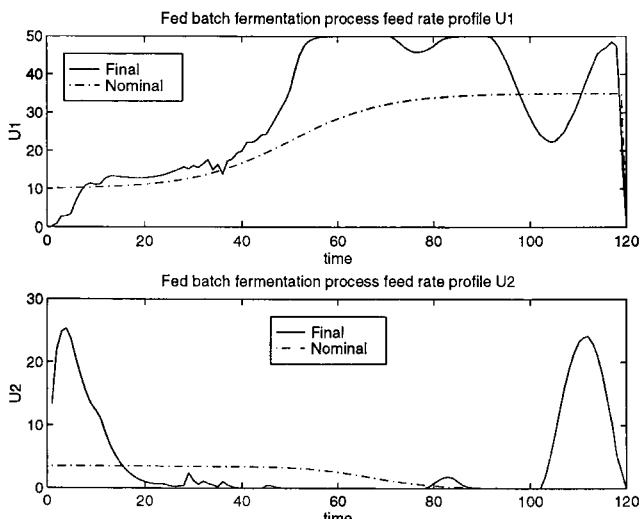


Fig 1 Nominal and final feed rate profiles

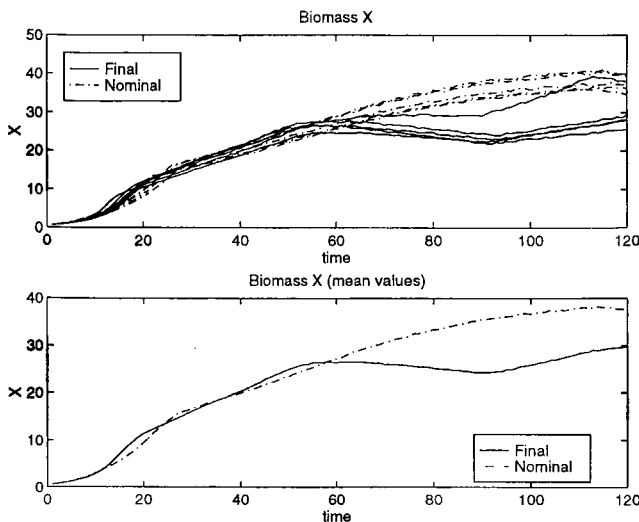


Fig 2 Biomass responses

subject to

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) + \alpha(k) \\ x(0) &= x_0 \end{aligned} \quad \dots (26)$$

where A and B are given above, r_1 is adjusted along the iterations of Algorithm 1, and x_0 is computed from the average initial value of the measured variables at every iteration. This average was computed by applying to the process the current input profile iterate $\{v(\cdot)\}$ five times. This implies that five batches were run for each iteration of Algorithm 1.

It is known to the authors (Becerra and Roberts, 1996b) that the use of Broyden's method for computing the dynamic first derivatives along the trajectories introduces sudden jumps in the resulting input profiles. In order to obtain smoother profiles the computed values of $v(k)$ obtained from Algorithm 1 were filtered through a first-order filter before being applied to the process:

$$\begin{aligned} v(k)^{\text{applied}} &= v(k-1)^{\text{Algorithm 1}} \\ &+ k_f(v(k)^{\text{Algorithm 1}} - v(k-1)^{\text{Algorithm 1}}) \end{aligned}$$

where the value of the constant k_f was 0.5.

The modified model-based problem (MMOP) was solved at each iteration of Algorithm 1 by using LQ techniques (Becerra, 1994; Becerra and Roberts, 1996a). The nominal feed profiles were used as initial guesses for starting the iterations of DISOPE, while the initial guess for the costate trajectories was $p^0(0) = [0 \ 0 \ 0 \ 0 \ 0]^T$, $k \in [0, 119]$. The initial guess for the derivative sequence to start Broyden's method iterations was $D(k) = [A \ B]$, $k \in [0, 118]$. Fig 1 shows the nominal and final (after 20 iterations) feed profiles. Two graphs are shown in each figure from Figs 2 to 8, each of these figures corresponding to a different variable. The upper graph in each figure shows the nominal and best (after 20 iterations) responses for several simulation runs, while the lower graph in each figure shows the average of nominal and best responses. Fig 7 shows that an improvement of 28.7% (as compared with the average nominal value) has been achieved in terms of the average of the end point value of the productivity measure $P(t_f)V(t_f)/t_f$, while a significant

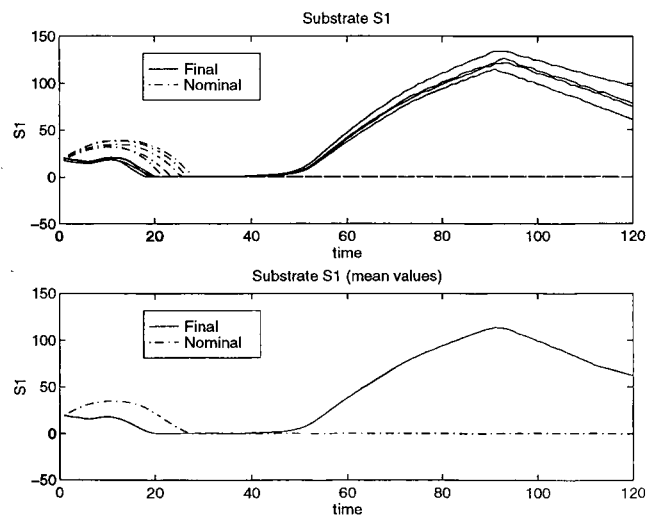


Fig 3 Substrate S_1 responses

improvement can also be clearly appreciated along most of the batch period. In Fig 8, an improvement of 28.7% may also be observed in the amount of product (Concentration $\times$ Volume) at the end of the batch period (as compared with the average nominal value).

Notice that the average final productivity measure achieves 90% of its end point value (89 837 units) at 80 time units. On the other hand, the average productivity measure for the nominal profiles is just 27 662 at 80 time units. Thus, if the batch period was reduced from 120 time units to 80 time units, an improvement of 224% in the average productivity measure would be achieved with the final profile, as compared to the value of the average productivity measure at 80 time units. In batch processes a reduction in the batch period is often highly desirable since more batches may be run in a given period of time (days, weeks).

4. Critical comments and further research

The main limitations of the method presented in this paper and where further research is required are:

- (a) It is assumed that measurements or estimates of the state variables of the process may be obtained after

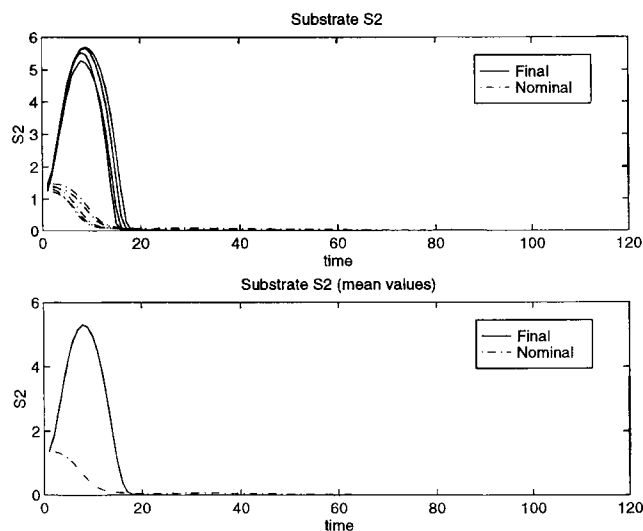


Fig 4 Substrate S_2 responses

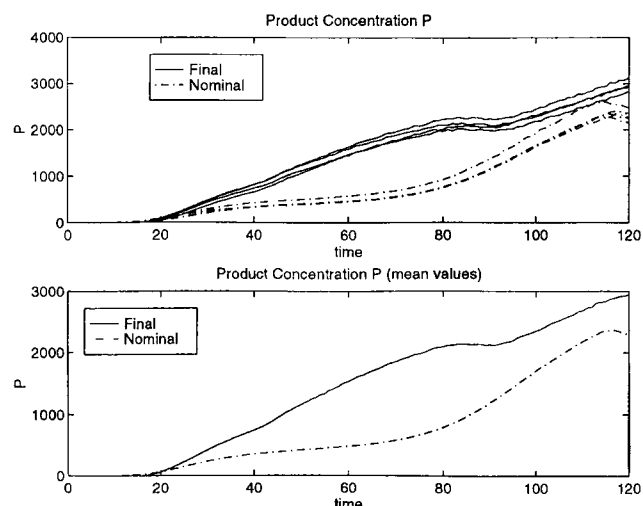


Fig 5 Concentration responses

the end of each batch. This is a characteristic of the method because it is formulated on the basis of the state space approach for optimal control. Given the values of the independent input trajectories along a

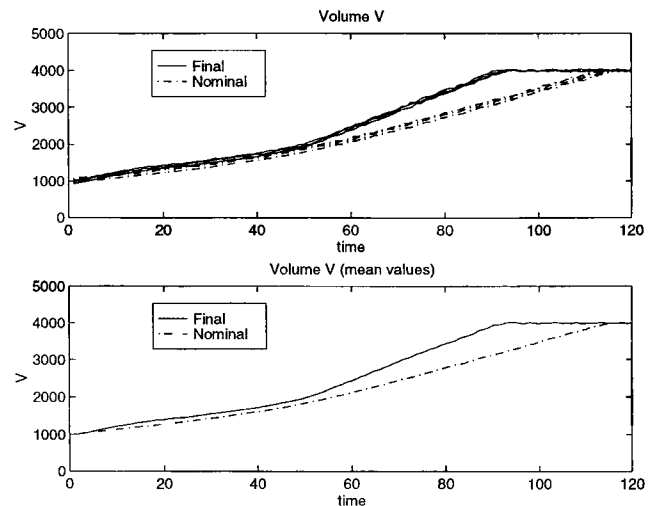


Fig 6 Volume responses

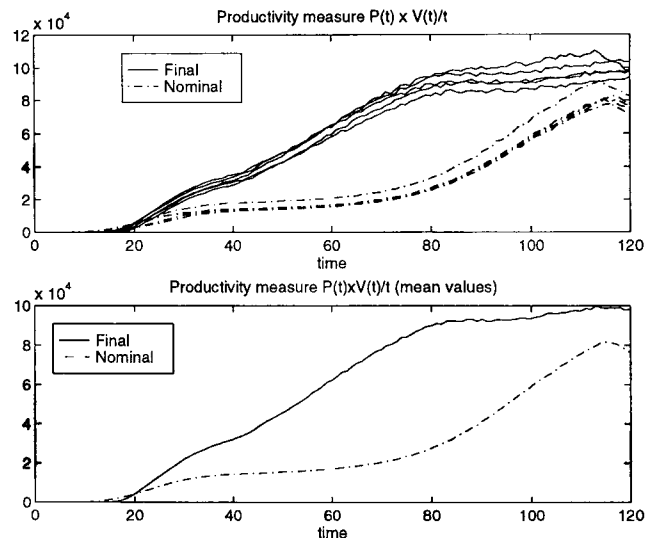


Fig 7 Productivity measure

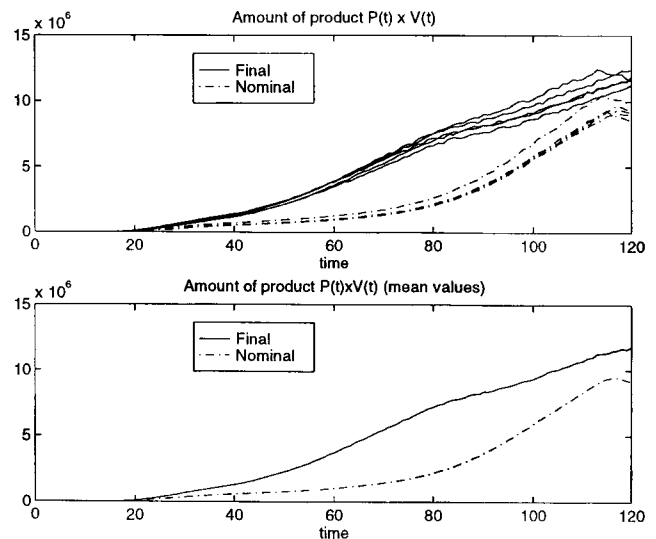


Fig 8 Amount of product

period of time, the state variables are defined as those variables which are required at an initial time to predict the future behaviour of the process along the same period of time. If not all the states are measured then the method may still be applied, but performance may be sub-optimal.

- (b) The estimation of derivatives is approximate. Broyden's method gives reasonable but approximate values of the dynamic first derivatives required and this has an effect on the final solution. The method also introduces sudden jumps along the input trajectories which had to be filtered out to approximate implementable profiles.
- (c) The calculation of the mean values of the state trajectories is carried out using just a finite number of batches. Furthermore, the proposed way of handling of batch-to-batch changes on the initial conditions and on the process response for the same input assumes that these random changes have a constant mean value.
- (d) The method for handling of state constraints proposed in this paper does not use feedback information, and hence state constraints are not enforced on-line.

5. Conclusions

An algorithm for Dynamic Integrated System Optimisation and Parameter Estimation has been formulated and applied to the dynamic optimisation of a benchmark fed-batch fermentation process. The iterations of DISOPE are integrated with the batchwise operation of the process. An extension of Broyden's method for computing the first derivatives required by DISOPE has been used. An averaging technique has been applied to optimise the process in spite of batch-to-batch variations in its dynamic behaviour and random changes of the initial conditions from batch to batch. The use of the techniques presented in the paper enables the batch-to-batch approximation of the dynamic optimum of the batch process in spite of deficiencies in the dynamic model used for the optimisation calculations. The results obtained indicate that the best feed profiles obtained using the DISOPE algorithm improve the productivity measure and amount of final product in 28.7%, as compared with the values corresponding to the nominal feed profiles. It has also been shown how the batch period may be reduced and still have a significant improvement in the productivity measure.

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